



11634 - Some like it hot! Mapping the inner 10 AU of embedded protoplanetary disks during their mineral condensation epoch.

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Melissa McClure (PI) (ESA Member)	Universiteit Leiden
Dr. Merel van 't Hoff (CoI) (US Admin CoI)	Purdue University
Dr. Edwin Anthony Bergin (CoI)	University of Michigan
Dr. John Tobin (CoI)	Associated Universities, Inc.
Dr. Sam Federman (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Capodimonte
Dr. Tom Megeath (CoI)	University of Toledo
Ms. Femke Ballieux (CoI) (ESA Member)	Universiteit Leiden
Mulan Madden (CoI) (ESA Member)	Universiteit Leiden
Savio Barros Oliveira (CoI)	University of Toledo

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MIRI				
	3	HOPS-221_MIRI	MIRI Medium Resolution Spectroscopy	(3) HOPS-221
	4	HOPS-166_MIRI	MIRI Medium Resolution Spectroscopy	(4) HOPS-166
NIRSpec				
	7	HOPS-166_NIRSpec	NIRSpec Fixed Slit Spectroscopy	(4) HOPS-166
	8	HOPS-221_NIRSpec	NIRSpec Fixed Slit Spectroscopy	(3) HOPS-221

ABSTRACT

JWST Proposal 11634 (Created: Friday, March 13, 2026, 4:09:00PM Eastern Standard Time) - Overview

The oldest solids in the Solar System condensed out of a high temperature gas phase reservoir, setting the $t=0$ moment for the Solar System's formation. Over time this gas reservoir cooled, and its size and location as a function of time limited the formation location of the terrestrial planets in the Solar System. The minerals condensing from this reservoir may have contributed as much as 96% of the mass of Earth, but this is still highly debated. This condensation sequence had never been observed in disks, until JWST detected a massive, warm SiO gas reservoir and condensed crystalline minerals at 1 AU in the embedded disk around an exceptional 0.1 Myr protostar, HOPS 315. A second protostar, HOPS 68, has been discovered by JWST to have the same SiO gas and crystalline silicate feature, suggesting a common origin, likely related to a spatially extended accretion heated disk midplane. These two protostars, along with two others, HOPS 221 and HOPS 166, showed strong crystalline silicate absorption with Spitzer. These four protostars are at different points in the 0.3 Myr condensation epoch seen in the meteoritic record. We request a modest 5.3 hours to observe HOPS 221 and HOPS 166 to detect the same copious gas phase absorption lines seen towards the two archival protostars. Analysis of the combined GO+AR dataset will provide detailed temperature, density, and gas/dust composition maps across the full 0.3 Myr epoch of mineral condensation for the first time. With these observations, we can begin to reveal the structure of planetary nurseries around other stars and connect them to the history of our own Solar System.

OBSERVING DESCRIPTION

We will observe 2 embedded disks with the following filters:

NIRSpec Fixed Slit. The targets are expected to be point sources, except for H₂ jets, which are not our primary science case. We use TA in WATA mode to place the targets in the slit and observe them with the G140H, G235H, and G395H filters. We dither with a 2-position nodding pattern, using the opposite nod for sky subtraction. Fixed Slit observations do not experience leakage through the MSA failed open shutters and require no additional calibrations.

We selected the subarray and readout patterns based on the brightness of our targets in the G395H filter and maintained the subarray for all other filters. HOPS 166 and HOPS 221, are bright enough to require the Subarray SUBS200A1 with the NRSRAPID readout mode to avoid saturation.

We aim for a $S/N \geq 100$ at 1.07 microns and 1.8-2.0 microns, and $S/N \sim 400$ at 4.1 microns.

MIRI

We use a 4-position dither pattern and use a FAST readout to minimize overheads. We do TA to enable optimal fringe correction. We take off-source simultaneous imaging to confirm the pointing of these observations and maximize bonus-science from any serendipitously observed sources. Our

JWST Proposal 11634 (Created: Friday, March 13, 2026, 4:09:00PM Eastern Standard Time) - Overview

S/N is >300 at all subchannels up to 22 microns. These targets are bright enough that the additional background subtraction provides little benefit.

Both MIRI and NIRSpec must be observed in a non-interruptible sequence for each target, to obtain our target emission lines quasi-simultaneously for these variable targets.

Proposal 11634 - Targets - Some like it hot! Mapping the inner 10 AU of embedded protoplanetary disks during their mineral condensa...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(3)	HOPS-221	RA: 05 42 47.0670 (85.6961125d) Dec: -08 17 6.93 (-8.28526d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Eruptive variables, FU Orionis stars, Protoplanetary disks, Protostars] Extended=NO				
	(4)	HOPS-166	RA: 05 36 25.1310 (84.1047125d) Dec: -06 44 41.84 (-6.74496d) Equinox: J2000	Proper Motion RA: 1.158 mas/yr Proper Motion Dec: 0.188 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Eruptive variables, FU Orionis stars, Protoplanetary disks, Protostars] Extended=NO				

Proposal 11634 - Observation 3 - Some like it hot! Mapping the inner 10 AU of embedded protoplanetary disks during their mineral co...

Observation	Proposal 11634, Observation 3: HOPS-221_MIRI												Fri Mar 13 21:09:00 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	HOPS-221	RA: 05 42 47.0670 (85.6961125d) Dec: -08 17 6.93 (-8.28526d) Equinox: J2000				Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Star Description=[Eruptive variables, FU Orionis stars, Protoplanetary disks, Protostars] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	4	1	1	11.1	175215				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	175215
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
	2		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	2	MEDIUM(B)	MRSLONG		FASTR1	22	1	1	Dither 1	4	4	244.204	
	2	MEDIUM(B)	MRSSHORT		FASTR1	22	1	1	Dither 1	4	4	244.204	
	3		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	3	LONG(C)	MRSLONG		FASTR1	12	1	1	Dither 1	4	4	133.202	
	3	LONG(C)	MRSSHORT		FASTR1	12	1	1	Dither 1	4	4	133.202	

Special Requirements	Group Observations 3, 8, Non-interruptible
----------------------	--

Proposal 11634 - Observation 4 - Some like it hot! Mapping the inner 10 AU of embedded protoplanetary disks during their mineral co...

Observation	Proposal 11634, Observation 4: HOPS-166_MIRI												Fri Mar 13 21:09:00 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(4)	HOPS-166	RA: 05 36 25.1310 (84.1047125d)				Proper Motion RA: 1.158 mas/yr						
			Dec: -06 44 41.84 (-6.74496d)				Proper Motion Dec: 0.188 mas/yr						
			Equinox: J2000				Epoch of Position: 2000						
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Eruptive variables, FU Orionis stars, Protoplanetary disks, Protostars] Extended=NO											
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging		Imager Subarray		Grating Wheel Direction			
	FND	All MRS				YES		FULL		Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	175303
	1	SHORT(A)	MRSLONG		FASTR1	10	1	1	Dither 1	4	4	111.002	
	1	SHORT(A)	MRSSHORT		FASTR1	10	1	1	Dither 1	4	4	111.002	
	2		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	2	MEDIUM(B)	MRSLONG		FASTR1	8	1	1	Dither 1	4	4	88.801	
	2	MEDIUM(B)	MRSSHORT		FASTR1	8	1	1	Dither 1	4	4	88.801	
	3		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
	3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	

Special Requirements	Group Observations 4, 7, Non-interruptible
----------------------	--

Proposal 11634 - Observation 7 - Some like it hot! Mapping the inner 10 AU of embedded protoplanetary disks during their mineral co...

Observation	Proposal 11634, Observation 7: HOPS-166_NIRSpec										Fri Mar 13 21:09:00 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(4)	HOPS-166	RA: 05 36 25.1310 (84.1047125d)			Proper Motion RA: 1.158 mas/yr						
			Dec: -06 44 41.84 (-6.74496d)			Proper Motion Dec: 0.188 mas/yr						
			Equinox: J2000			Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Eruptive variables, FU Orionis stars, Protoplanetary disks, Protostars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	226203	
Template	HFF Readout Mode			Slit				Subarray				
	false			S200A1				SUBS200A1				
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	S200A1	NRSRAPID	3	1	1	NONE	4	4	25.01	226203
	2	G235H/F170LP	S200A1	NRSRAPID	4	1	2	NONE	4	4	31.242	226203
	3	G395H/F290LP	S200A1	NRSRAPID	15	1	3	NONE	4	4	99.794	226203

Special Requirements	Group Observations 4, 7, Non-interruptible
----------------------	--

Proposal 11634 - Observation 8 - Some like it hot! Mapping the inner 10 AU of embedded protoplanetary disks during their mineral co...

Observation	Proposal 11634, Observation 8: HOPS-221_NIRSpec										Fri Mar 13 21:09:00 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	HOPS-221	RA: 05 42 47.0670 (85.6961125d) Dec: -08 17 6.93 (-8.28526d) Equinox: J2000			Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	Category=Star											
	Description=[Eruptive variables, FU Orionis stars, Protoplanetary disks, Protostars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	226202	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				SUBS200A1			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	S200A1	NRSRAPID	15	1	1	NONE	4	4	99.794	
	2	G235H/F170LP	S200A1	NRSRAPID	50	1	2	NONE	4	4	317.914	
	3	G140H/F100LP	S200A1	NRSRAPID	400	1	3	NONE	4	4	2499.114	

Special Requirements	Group Observations 3, 8, Non-interruptible
----------------------	--